

Proof

Read the instruction word first — it tells you which of the four proofs to write. Every proof ends with a sentence.



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THE TOPIC ON ONE MAP

Proof

01 — Algebraic proof

$2n, 2n + 1$

consecutive

factor out

02 — Proof by exhaustion

odd / even

$3k, 3k + 1, 3k + 2$

a table of cases

03 — Counter-example

one example

show it fails

04 — Inequalities

complete the square

$(a - b)^2 \geq 0$

discriminant

THE DICTIONARY OF PROOF

even = $2n$ odd = $2n + 1$ consecutive = $n, n + 1$ multiple of k
 $= k(\dots)$

finish with a sentence: "... which is even, so the statement is true for all n "

01 Algebraic Proof: Write It in Symbols

Translate the words into algebra with the dictionary, simplify, then **factor out** the number you need. The last line is a sentence that repeats what you were asked to prove.

WHEN YOU SEE → YOU DO

"use algebra to prove", "for all integers n "

→ write in n , simplify, factor

"consecutive odd numbers"

→ $2n + 1, 2n + 3$

"is a multiple of 8"

→ get $8(\dots)$ — every term must share the 8

"is always even but never a multiple of 4"

→ $2(\text{odd})$: e.g. $4(\dots) + 2$

"prove C has no stationary points"

→ $\frac{dy}{dx}$ has negative discriminant, or is a square + positive

WORKED MODEL • JANUARY 2026

p and q are consecutive odd numbers, $p > q > 0$. Prove $p^2 - q^2$ is a multiple of 8.

1 $q = 2n + 1, p = 2n + 3$

dictionary

2 $p^2 - q^2 = (p - q)(p + q) = 2(4n + 4)$

difference of two squares

ANSWER $8(n + 1)$: a multiple of 8

× **TRAP** Using the same letter for two different odd numbers ($2n+1$ and $2n+1$). Consecutive odd numbers are $2n + 1$ and $2n + 3$.

IDEA 01

Practise: Algebraic Proof: Write It in Symbols

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** The dictionary

P2 • TOPIC 1 3 marks

Prove that the sum of two consecutive odd numbers is a multiple of 4.

Odd numbers: $2n + 1$ and $2n + 3$.

Q2 **DR ESLAM** Difference of squares

P2 • TOPIC 1 3 marks

Prove that the difference between the squares of two consecutive integers is always odd.

SELECTED PRACTICE

Q3 **PRACTICE** Practice · Easy

P2 · TOPIC 1 1 mark

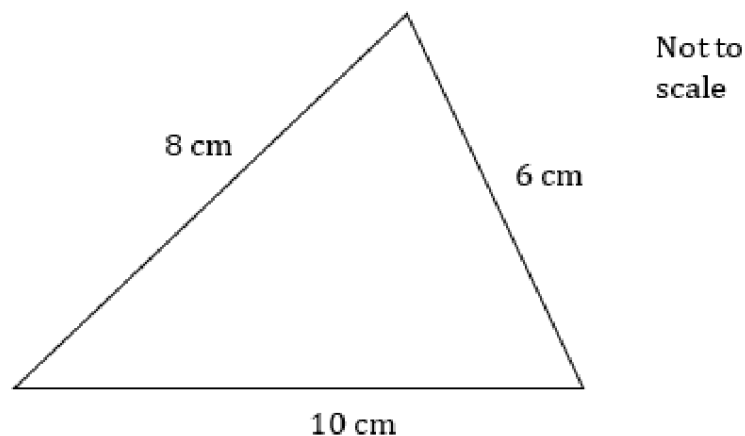
- 1 In a mathematical argument, how are three consecutive integers usually denoted algebraically?

(1 mark)

Q4 **PRACTICE** Practice · Medium

P2 · TOPIC 1 4 marks

- 9 Prove that a triangle with side lengths of 8 cm, 6 cm and 10 cm must contain a right-angle. You may use the diagram below to help.



(4 marks)

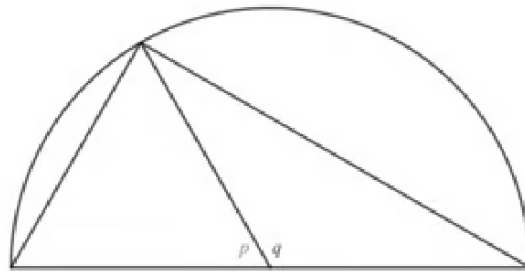
Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 1 4 marks

- 9 Prove the angle at the circumference in a semi-circle is a right angle.

You may use the diagram below to help.



Not to
scale

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · June 2021 · Q3

P2 · TOPIC 1 5 marks

3. (i) Prove that for all single digit prime numbers, p ,

$$p^3 + p \text{ is a multiple of } 10 \quad (2)$$

- (ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of } 3 \quad (3)$$

(Total 5 marks)

8. (i) A student states

“If x and y are irrational numbers, $x \neq y$, then xy is also irrational.”

Show, by counter example, that this statement is not always true.

(1)

(ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

(4)

(Total for Question 8 is 5 marks)

Q8 PAST PAPER Pearson IAL WMA12 · January 2026 · Q6

P2 · TOPIC 1 8 marks

6. (i) Given that p and q are consecutive odd numbers, where $p > q > 0$, prove that

$$p^2 - q^2$$

is a multiple of 8

(4)

- (ii) The curve C has equation

$$y = x^3 + 12x^2 + 49x + 2$$

Prove that C has no stationary points.

(4)

(Total for Question 6 is 8 marks)

02 Proof by Exhaustion: Every Case

Split *all* the possibilities into a small number of cases, and prove the statement in each one. Say at the start why your cases cover everything, and at the end that every case works.

WHEN YOU SEE → YOU DO

“use proof by exhaustion”, “if n is an integer”

→ cases $n = 2k$ and $n = 2k + 1$

“not divisible by 3”

→ cases $3k$, $3k + 1$, $3k + 2$ — show the remainder is never 0

“not a multiple of 4”

→ cases $4k$, $4k + 1$, $4k + 2$, $4k + 3$

“integers greater than 0 such that $a + b + c = 10$ ”

→ list every triple in a table

“for all single digit prime numbers”

→ check 2, 3, 5, 7 one by one

ZONE PARITY

2 cases

$2k$ $2k + 1$

ZONE MOD 3

3 cases

$3k$ $3k + 1$ $3k + 2$

ZONE A LIST

finite set

a table: check each row

WORKED MODEL • JANUARY 2023

Prove that $2n^2 + n + 1$ is not divisible by 3. Case $n = 3k + 1$:

$$1 \quad 2(9k^2 + 6k + 1) + (3k + 1) + 1 = 18k^2 + 15k + 4$$

substitute

$$2 \quad = 3(6k^2 + 5k + 1) + 1$$

a multiple of 3, plus 1

ANSWER remainder 1, so not divisible by 3

× **TRAP** Forgetting a case, or stopping without the concluding sentence “all cases are covered”.

IDEA 02

Practise: Proof by Exhaustion: Every Case

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Odd and even cases

P2 · TOPIC 1 4 marks

Prove that $n^2 + n$ is even for every integer n .

Case $n = 2k$ and case $n = 2k + 1$.

Q10 DR ESLAM A small list

P2 · TOPIC 1 3 marks

Prove by exhaustion that $n^2 + 1$ is not divisible by 3 for $n = 1, 2, 3, 4, 5$.

SELECTED PRACTICE

Q11

PRACTICE

Practice · Easy

P2 · TOPIC 1

2 marks

7 Prove by exhausting all possible factors that 11 is a prime number.

(2 marks)**Q12**

PRACTICE

Practice · Hard

P2 · TOPIC 1

4 marks

4 The set of numbers S is defined as all positive integers greater than 5 and less than 10.

Prove by exhaustion that the square of all values in S differ from a multiple of 5 by 1.

(4 marks)

Q13**PRACTICE**

Practice · Very Hard

P2 · TOPIC 1 6 marks

- 10** A standard chess board has 64 1×1 - sized squares.
It also has 1 8×8 - sized square.

Prove that there are 204 squares on a standard chess board.

(6 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA12 · January 2022 · Q10

P2 · TOPIC 1 5 marks

10. (i) Prove by counter example that the statement

“if p is a prime number then $2p + 1$ is also a prime number”

is not true.

(1)

- (ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

Q15 **PAST PAPER** Pearson IAL WMA12 · October 2022 · Q1**P2 · TOPIC 1** **3 marks**

1. Given that a , b and c are integers greater than 0 such that

- $c = b + 2$
- $a + b + c = 10$

Prove, by exhaustion, that the product of a , b and c is always even.

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

a	b	c
	1	
	2	

(Total for Question 1 is 3 marks)

Q16 PAST PAPER Pearson IAL WMA12 · January 2023 · Q10

P2 · TOPIC 1 4 marks

10. A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the box below.

Consider the case when $n = 3k$

$$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$

which is not divisible by 3

Complete this proof.

(4)

(Total for Question 10 is 4 marks)

Q17 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q8**P2 · TOPIC 1** **5 marks**

8. (i) A student writes the following statement:

“When a and b are consecutive **prime** numbers, $a^2 + b^2$ is never a multiple of 10”

Prove by counter example that this statement is **not** true.

(2)

- (ii) Given that x and y are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

(3)

(Total for Question 8 is 5 marks)

Q18 **PAST PAPER** Pearson IAL WMA12 · October 2023 · Q1**P2 · TOPIC 1** **3 marks**

1. Given that a , b and c are integers greater than 0 such that

- $c = 3a + 1$
- $a + b + c = 15$

prove, by exhaustion, that the product abc is always a multiple of 4

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

	a	b	c	abc
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(Total for Question 1 is 3 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q8

P2 · TOPIC 1 6 marks

8. (i) Use a counter example to show that the following statement is **false**

“ $n^2 + 3n + 1$ is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Use algebra to prove by exhaustion that for all $n \in \mathbb{N}$

“ $n^2 - 2$ is **not** a multiple of 4”

(4)

(Total for Question 8 is 6 marks)

Q20 **PAST PAPER** Pearson IAL WMA12 · October 2024 · Q11**P2 · TOPIC 1** **6 marks**

11. (i) Prove by counter example that the statement

“If n is a prime number then $3^n + 2$ is also a prime number.”

is false.

(2)

(ii) Use proof by exhaustion to prove that if m is an integer that is **not** divisible by 3, then

$$m^2 - 1$$

is divisible by 3

(4)

(Total for Question 11 is 6 marks)

Q21**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q10

P2 · TOPIC 1**4 marks****10:****In this question you must show detailed reasoning.**

Use algebra to prove by exhaustion that,

for all positive integers m that are **not** multiples of 3, the value of

$$m^2 + 3m + 2$$

is always a multiple of 3

(4)**(Total for Question 10 is 4 marks)**

03 Disproof by Counter-Example

To show a statement is **false**, one example is enough. Choose the value, work it out, and say clearly why it breaks the statement.

WHEN YOU SEE → YOU DO

“show that the statement is not true”

→ one value that fails

“is prime for all n ”

→ try small n until you find a composite, show its factors

“is a perfect cube for all n ”

→ find a value between two cubes

“for all real x ” with a square root or a sign

→ try a negative number

“If x and y are irrational then xy is irrational”

→ $\sqrt{2} \times \frac{1}{\sqrt{2}} = 1$

WORKED MODEL • JUNE 2022

Show that “ $(n + 1)^3 - n^3$ is prime for all $n \in \mathbb{N}$ ” is false.

1 $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$

simplify

2 $n = 5: 75 + 15 + 1 = 91$

try values

ANSWER $91 = 7 \times 13$, not prime

× **TRAP** Giving the example without the reason. Write “ $91 = 7 \times 13$, so it is not prime”.

IDEA 03

Practise: Disproof by Counter-Example

My questions first, then selected practice, then real past papers.

Q22 DR ESLAM One example is enough

P2 · TOPIC 1 2 marks

Show that the statement “ $n^2 - n + 11$ is prime for all positive integers n ” is false.

SELECTED PRACTICE**Q23** PRACTICE Practice · Easy

P2 · TOPIC 1 2 marks

6 Use a counter-example to show that $\sqrt{x^2} \neq x$.

(2 marks)

Q24**PRACTICE**

Practice · Hard

P2 · TOPIC 1**2 marks**

- 5 Use a counter-example to prove that not all integers of the form $2^n - 1$, where n is an integer, are prime.

(2 marks)**Q25****PRACTICE**

Practice · Very Hard

P2 · TOPIC 1**2 marks**

- 5 Give an example to show when the following statement is both true and false.

The square of a positive integer is always greater than doubling it.

(2 marks)

PAST PAPERS

Q26 PAST PAPER Pearson IAL WMA12 · June 2022 · Q3

P2 · TOPIC 1 7 marks

3. (i) Show that the following statement is **false**:

$$“(n + 1)^3 - n^3 \text{ is prime for all } n \in \mathbb{N}” \quad (2)$$

- (ii) Given that the points $A(1, 0)$, $B(3, -10)$ and $C(7, -6)$ lie on a circle, prove that AB is a diameter of this circle. (5)

(Total 7 marks)

04 Proving Inequalities: Complete the Square

Move everything to one side so you need to show “something > 0 ” or “ ≥ 0 ”. Then complete the square: a square is never negative. For two variables, use $(a - b)^2 \geq 0$.

WHEN YOU SEE → YOU DO

“prove $(x + 1)^2 > 6x - 4$ for all real x ”

→ one side: $x^2 - 4x + 5 = (x - 2)^2 + 1 > 0$

“ $\frac{x+y}{2} \geq \sqrt{xy}$ ”

→ $(\sqrt{x} - \sqrt{y})^2 \geq 0$, then expand

“ $3x + 1 \geq 2\sqrt{3x}$ ”

→ $(\sqrt{3x} - 1)^2 \geq 0$

“for all positive x and y ”

→ square roots are allowed only because they are positive

“show that it is not true for all real numbers”

→ counter-example with a negative

Everything on one side

Complete the square

Square ≥ 0

Conclusion

WORKED MODEL • JUNE 2026

Prove that $(x + 1)^2 > 6x - 4$ for all real x .

1 $x^2 + 2x + 1 - 6x + 4 = x^2 - 4x + 5$

one side

2 $= (x - 2)^2 + 1$

complete the square

ANSWER $(x - 2)^2 \geq 0$, so $(x - 2)^2 + 1 \geq 1 > 0$

IDEA 04

Practise: Proving Inequalities: Complete the Square

My questions first, then selected practice, then real past papers.

Q27 **DR ESLAM** Complete the square

P2 · TOPIC 1 3 marks

Prove that $x^2 - 6x + 10 > 0$ for all real x .

Q28 **DR ESLAM** Two variables

P2 · TOPIC 1 3 marks

Prove that $a^2 + b^2 \geq 2ab$ for all real a and b .

SELECTED PRACTICE

Q29

PRACTICE

Practice · Easy

P2 · TOPIC 1 1 mark

4 Explain why $(x - 3)^2 \geq 0$ for all real values of x .

(1 mark)

Q30

PRACTICE

Practice · Medium

P2 · TOPIC 1 2 marks

2 Prove that $x^2 + 2 \geq 2$ for all values of x .

(2 marks)

Q31**PRACTICE**

Practice · Very Hard

P2 · TOPIC 1 4 marks

7 Prove that, if n is negative, $n^4 - n^3$ is positive.

(4 marks)

PAST PAPERS

Q32**PAST PAPER**

Pearson IAL WMA12 · January 2021 · Q5

P2 · TOPIC 1 4 marks

5. (i) Use algebra to prove that for all $x \geq 0$

$$3x + 1 \geq 2\sqrt{3x} \quad (3)$$

(ii) Show that the following statement is not true.

“The sum of three consecutive prime numbers is always a multiple of 5”
(1)

(Total 4 marks)

Q33**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q9

P2 · TOPIC 1**4 marks**

9. (a) Prove that for all positive values of x and y ,

$$\frac{x + y}{2} \geq \sqrt{xy} \quad (3)$$

- (b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

(Total 4 marks)

Q34 **PAST PAPER** Pearson IAL WMA12 · June 2024 · Q5**P2 · TOPIC 1** **6 marks**

5. **In this question you must show detailed reasoning.**

- (a) Given that x and y are positive numbers such that

$$(x - y)^3 > x^3 - y^3$$

prove that

$$y > x$$

(4)

- (b) Using a counter example, show that the result in part (a) is not true for all real numbers.

(2)

(Total for Question 5 is 6 marks)

Q35**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q4

P2 · TOPIC 1 6 marks

4. **In this question you must show detailed reasoning.**

(i) Show by counterexample that the statement

$$\text{“}f(n) = \frac{5}{2}n^3 - \frac{5}{2}n^2 - 19n + 20 \text{ is a perfect cube for all positive integers } n\text{”}$$

is **not** true.

(3)

(ii) Prove that the following statement is true

$$\text{“}(x + 1)^2 > 6x - 4 \text{ for all real } x\text{”}$$

(3)

(Total for Question 4 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|---|
| 01 | Algebraic | Dictionary; simplify; factor out; sentence. |
| 02 | Exhaustion | Cases that cover everything; prove each; “all cases covered”. |
| 03 | Counter-example | One value; show exactly why it fails. |
| 04 | Inequalities | One side; complete the square; a square ≥ 0 . |

BOTTOM LINE Read the instruction word → choose the proof → write it in symbols → end with a sentence.

Answers

Proof • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ALGEBRAIC PROOF: WRITE IT IN SYMBOLS

- Q1** Dr Eslam • The dictionary $(2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$, a multiple of 4
- Q2** Dr Eslam • Difference of squares $(n + 1)^2 - n^2 = 2n + 1$, which is odd
- Q3** Practice • Easy $n, n + 1, n + 2$
- Q4** Practice • Medium $6^2 + 8^2 = 100 = 10^2$, so by the converse of Pythagoras the angle opposite the 10 cm side is 90°
- Q5** Practice • Very Hard join the centre to the point on the circle: two isosceles triangles, $2\alpha + 2\beta = 180^\circ$, so $\alpha + \beta = 90^\circ$
- Q6** Pearson IAL WMA12 • June 2021 • Q3 (i) Proven by checking $p = 2, 3, 5, 7$. (ii) Proven: $(n + 1)^3 - n^3$ leaves remainder 1 on division by 3
- Q7** Pearson IAL WMA12 • January 2025 • Q8 (i) For example, $x = \sqrt{2}$ and $y = \frac{1}{\sqrt{2}}$ give $xy = 1$. (ii) Proven
- Q8** Pearson IAL WMA12 • January 2026 • Q6 (i) Proven. (ii) C has no stationary points

IDEA 02 • PROOF BY EXHAUSTION: EVERY CASE

- Q9** Dr Eslam • Odd and even cases
 $n = 2k$: $4k^2 + 2k = 2(2k^2 + k)$; $n = 2k + 1$:
 $4k^2 + 6k + 2 = 2(2k^2 + 3k + 1)$ — even in both cases
- Q10** Dr Eslam • A small list
 2, 5, 10, 17, 26: none is a multiple of 3
- Q11** Practice • Easy
 11 is not divisible by 2 or 3 (the primes up to $\sqrt{11}$), so it is prime
- Q12** Practice • Hard
 $36 = 35 + 1$, $49 = 50 - 1$, $64 = 65 - 1$, $81 = 80 + 1$
- Q13** Practice • Very Hard
 there are $(9 - k)^2$ squares of size $k \times k$: $64 + 49 + 36 + 25 + 16 + 9 + 4 + 1 = 204$
- Q14** Pearson IAL WMA12 • January 2022 • Q10
 (i) For $p = 7$, $2p + 1 = 15$, not prime. (ii) Proven
- Q15** Pearson IAL WMA12 • October 2022 • Q1
 The possible triples are (6, 1, 3), (4, 2, 4) and (2, 3, 5), giving products 18, 32 and 30. Each product is even, so abc is always even
- Q16** Pearson IAL WMA12 • January 2023 • Q10
 For $n = 3k$, $n = 3k + 1$ and $n = 3k + 2$, the expression has remainders 1, 1 and 2 respectively when divided by 3. Therefore $2n^2 + n + 1$ is never divisible by 3
- Q17** Pearson IAL WMA12 • June 2023 • Q8
 (i) A counterexample is $a = 7$, $b = 11$, since $7^2 + 11^2 = 170$. (ii) Exhaustion proves the inequality for $(x, y) = (2, 2), (2, 4), (4, 2), (4, 4)$
- Q18** Pearson IAL WMA12 • October 2023 • Q1
 The only possible triples are (1, 10, 4), (2, 6, 7) and (3, 2, 10), giving products 40, 84 and 60. Each is a multiple of 4, so abc is always a multiple of 4
- Q19** Pearson IAL WMA12 • January 2024 • Q8
 (i) Counterexample: $n = 6$ gives 55, which is not prime. (ii) Proven by considering $n = 4k, 4k + 1, 4k + 2, 4k + 3$
- Q20** Pearson IAL WMA12 • October 2024 • Q11
 (i) Counterexample: $n = 5$ gives $3^5 + 2 = 245$, not prime. (ii) Proven
- Q21** Pearson IAL WMA12 • June 2025 • Q10
 Proven

IDEA 03 · DISPROOF BY COUNTER-EXAMPLE

- Q22** Dr Eslam · One example is enough
 $n = 11: 121 = 11^2$, not prime
- Q23** Practice · Easy
 $x = -2: \sqrt{(-2)^2} = 2 \neq -2$
- Q24** Practice · Hard
 $n = 4: 2^4 - 1 = 15 = 3 \times 5$, not prime
- Q25** Practice · Very Hard
 true for $n = 3$ ($9 > 6$); false for $n = 1$ ($1 < 2$) or $n = 2$ ($4 = 4$)
- Q26** Pearson IAL WMA12 · June 2022 · Q3
 (i) Counterexample: $n = 5$ gives 91, not prime. (ii) Proven: AB is a diameter

IDEA 04 · PROVING INEQUALITIES: COMPLETE THE SQUARE

- Q27** Dr Eslam · Complete the square
 $(x - 3)^2 + 1 \geq 1 > 0$
- Q28** Dr Eslam · Two variables
 $a^2 + b^2 - 2ab = (a - b)^2 \geq 0$
- Q29** Practice · Easy
 a square is never negative, so $(x - 3)^2 \geq 0$
- Q30** Practice · Medium
 $x^2 \geq 0$, so $x^2 + 2 \geq 2$
- Q31** Practice · Very Hard
 $n^4 - n^3 = n^3(n - 1)$; for $n < 0$ both $n^3 < 0$ and $n - 1 < 0$, so the product is positive
- Q32** Pearson IAL WMA12 · January 2021 · Q5
 (i) Proven. (ii) For example, $5 + 7 + 11 = 23$, which is not a multiple of 5
- Q33** Pearson IAL WMA12 · October 2021 · Q9
 (a) Proven. (b) For example, $x = y = -1$ gives $-1 \not\geq 1$
- Q34** Pearson IAL WMA12 · June 2024 · Q5
 (a) Proven. (b) For example, $x = 1, y = -1$ satisfies the inequality but not $y > x$
- Q35** Pearson IAL WMA12 · June 2026 · Q4
 (i) $f(5) = 175$; $5^3 = 125 < 175 < 216 = 6^3$, so not a cube (ii) $(x+1)^2 - (6x-4) = (x-2)^2 + 1 > 0$